Deep Learning Capstone Project Report

1. Introduction

This capstone project focuses on implementing and analyzing various deep learning architectures and techniques learned during the course Deep Learning.

The project includes:

- Feedforward Neural Network (FNN) for classification
- Convolutional Neural Network (CNN) for image classification
- Long Short-Term Memory (LSTM) for sequence modeling

These experiments demonstrate the practical applications of deep learning in different domains.

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2. Methodology

Experiment 1: Feedforward Neural Network

Dataset Used: UCI Adult Income Dataset

Preprocessing Steps:

- Missing values handled
- Categorical values encoded
- Features normalized
- Train-test split applied

Model Architecture:

- Input Layer
- Dense Layer (128 neurons, ReLU)
- Dropout Layer
- Dense Layer (64 neurons, ReLU)
- Output Layer (Sigmoid)

Experiment 2: CNN for Image Classification

Dataset Used: Fashion-MNIST

Preprocessing:

- Normalized pixel values
- Reshaped images

Model Architecture:

- Conv2D + MaxPooling
- Conv2D + MaxPooling
- Flatten
- Dense + Dropout
- Output Layer (Softmax)

Experiment 3: LSTM for Sequence Modeling

Dataset Used: Stock Price Dataset

Preprocessing:

- Closing price selected
- Normalization
- Sequence windowing

Model Architecture:

- LSTM Layer
- LSTM Layer
- Dense Output Layer

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3. Implementation Details

Framework Used: TensorFlow / Keras

Hyperparameter	Value
Optimizer	Adam
Epochs	10–20
Batch Size	32
Loss Function	CrossEntropy / MSE

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4. Results

Experiment 1:

- Accuracy achieved: 85% approx

Experiment 2:

- Accuracy achieved: 88% approx
- Precision, Recall, F1-score calculated
- Confusion Matrix generated

Experiment 3:

- RMSE calculated for prediction performance
- Actual vs Predicted graph generated

Graphs included:

- Training vs Validation Accuracy
- Training vs Validation Loss

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5. Discussion

Observations:

- FNN works well for structured/tabular data
- CNN performs effectively on image classification
- LSTM captures temporal dependencies in sequence data

Challenges Faced:

- Overfitting in CNN
- Data preprocessing complexity
- Hyperparameter tuning

Comparison:

Model	Best For
FNN	Tabular Data

Model	Best For
CNN	Images
LSTM	Time-Series / Sequential Data

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6. Conclusion

This project provided hands-on experience with multiple deep learning architectures.

Key learnings:

- Designing neural networks
- Model evaluation techniques
- Handling overfitting
- Real-world applicability of deep learning

Deep learning models can be effectively applied in finance, healthcare, image recognition, and forecasting systems.

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7. References

- UCI Machine Learning Repository
- TensorFlow Documentation
- Keras Documentation
- Research papers and online tutorials